# Methodology for Wide Band-Gap Device Dynamic Characterization

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***Abstract*—The double pulse test (DPT) is a widely accepted method to evaluate the dynamic behavior of power devices. Considering the high switching-speed capability of wide band-gap devices, the test results are very sensitive to the alignment of voltage and current (V–I) measurements. Also, because of the shoot-through current induced by *C*dv/dt (i.e., cross-talk), the switching losses of the nonoperating switch device in a phase-leg must be considered in addition to the operating device. This paper summarizes the key issues of the DPT, including components and layout design, measurement considerations, grounding effects, and data processing. Additionally, a practical method is proposed for phase-leg switching loss evaluation by calculating the difference between the input energy supplied by a dc capacitor and the output energy stored in a load inductor. Based on a phase-leg power module built with 1200-V/50-A SiC MOSFETs, the test results show that this method can accurately evaluate the switching loss of both the upper and lower switches by detecting only one switching current and voltage, and it is immune to V–I timing misalignment errors.**

***Index Terms*—Double pulse test (DPT), dynamic characterization, phase-leg configuration wide bandgap.**

## I. Introduction

THE double pulse test (DPT) is a widely accepted method to assess the dynamic performance of power devices [1]–[9]. Two pulses are sent to the device under test (DUT) in a clamped inductive load circuit, as shown in Fig. 1. By regulating the dc bus voltage and first pulse duration, the DUT's switching transients can be captured under any desired voltage and current (V–I) conditions at the end of the first pulse and beginning of the second pulse. Since the DUT switches only twice for each test, the device junction temperature rise due to the switching loss is negligibly small. DPT results quantify the switching performance of the power devices and provide a basis for converter design, such as switching frequency and dead-time selections, thermal management, and efficiency estimation [1]–[9].

The DPT setup is illustrated in Fig. 2. For the control stage, the double-pulse signals with adjustable pulse widths are sent from the microcontroller or function generator, which is controlled by a personal computer (PC). The switching waveforms are measured by an oscilloscope, and then the data are transmitted back to the PC. The PC is responsible for programming

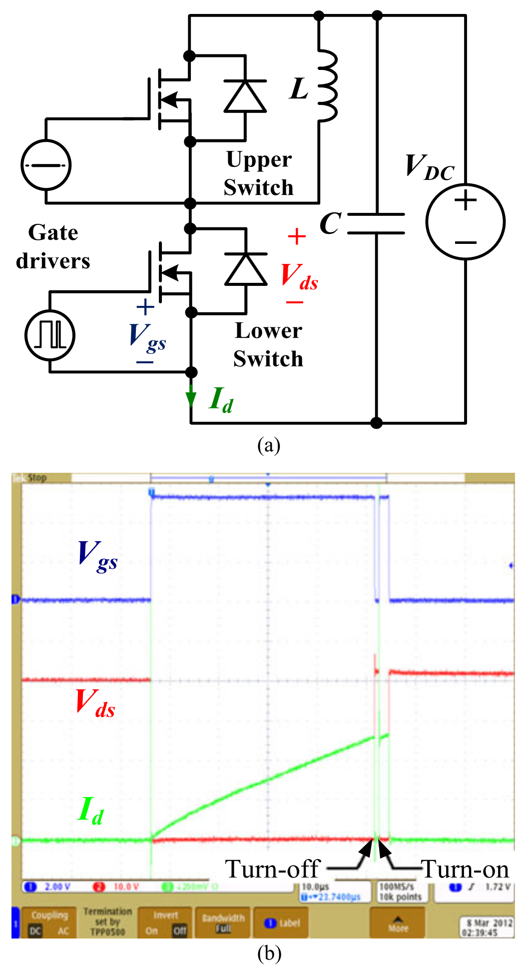


Fig. 1. DPT circuit: (a) schematics of DPT and (b) typical waveforms of DPT.

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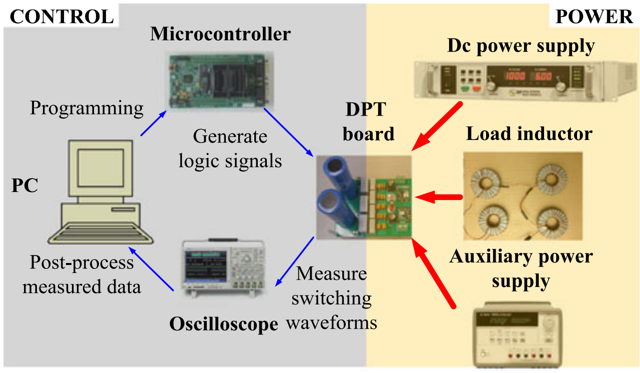


Fig. 2. DPT bench setup.

the double-pulse signals, collecting switching waveforms, and postprocessing the raw data. The DPT board includes a dc capacitor bank, gate driver circuits, power devices, and peripheral interfaces to connect with a dc voltage supply, load inductor, and auxiliary power supply. Because of their high switching-speed capability, the challenge for switching characterization of WBG devices primarily lies in the DPT board layout, precise measurement of switching waveforms, and data processing.

The physical layout considerations of a DPT board for SiC JFETs were presented in [2]. However, this optimized layout for the transistor−diode pair is not easily achieved in a phase-leg configuration, because it includes additional components. More instructions to minimize parasitic inductance in the switching loop of WBG-based voltage source converter were discussed in [10]–[12]. Some of these concepts are applicable to the DPT board, but many cannot be used directly due to the current sensor that is inserted in the switching loop.

The measurement issues of fast switching transients were discussed comprehensively in [5], [6], [13]–[16], including bandwidth requirements, voltage and current probe selection, and the alignment of voltage and current measurements. However, measurement techniques have continued to improve, and the probes summarized and recommended in [5] should be updated. Also, more attention should be paid to V−I alignment techniques due to the high switching-speed capability of WBG devices.

Data processing is also critical for switching characterization [5], [14]. Commonly, during the switching transient of one device in a phase-leg, the switching losses of its complementary device are neglected. However, the fast-switching WBG devices may induce more serious interference between two devices in a phase-leg (i.e., cross-talk), potentially leading to shoot-through current losses [17]– [24]. The Miller capacitance $C_{gd}$ couples the power loop *dv/dt* to the gate loop of the nonoperating device, and the resulting gate voltage spike may be sufficient to turn on the nonoperating device channel and result in a shoot-through current. Consequently, the conventional way of data processing, which only focuses on the switching loss of the operating device, must be re-evaluated.

This paper focuses on the phase-leg configuration, which is the basic cell of one of the most widely used converter types, the voltage source converter, and aims at presenting a methodology for dynamic characterization of WBG devices, including knowledge gained in previous studies as well as the experiences accumulated from new experiments. First, the design criteria of DPT circuits are described, including components (e.g., load inductor, dc capacitor bank) and physical layout. Second, switching voltage and current measurement requirements, the latest probes, and V-I alignment techniques are reviewed. Third, the impact of probe grounding effects on the measurement of switching waveforms is illustrated. Fourth, the mechanism causing additional switching losses by cross-talk are discussed, and a practical method of data processing is proposed for the switching loss assessment. Finally, a double pulse tester with 1200-V/50-A SiC MOSFETs in a phase-leg power module is established for experimental verification.

## II. Design Criteria of the DPT Circuit

As shown in Fig. 1(a), in addition to the DUT, the DPT circuit consists of a load inductor, dc capacitor, and gate driver circuits. The design of these components and their physical interconnections significantly affect the switching behavior of the DUT.

### A. Design Criteria of Main Components

The load inductor is used to establish the desired current during the first pulse and keep this current nearly constant during the following turn-off and turn-on transients. Therefore, the selected load inductance $L$ should be large enough to limit the inductive current variation $\Delta I_L$ during the switching transient yielding

$$L \geq \frac{V_{\rm DC}}{\Delta I_L} t_{\rm sw} = \frac{V_{\rm DC}}{k_i \times I_L} t_{\rm sw} \qquad (1)$$

where $t_{\rm sw}$ is the time to complete a switching event, whose preliminary value can be obtained from the device's datasheet, $k_i$ represents the current ripple coefficient, which is practically selected to be 1– 5%, and $V_{\rm DC}$ and $I_{\rm L}$ refer to the operating voltage and current. Usually, the switching performance of the DUT is characterized under several different operating points. According to (1), the worst condition for load inductance selection is under the maximum operating voltage and minimum current. According to this required inductance along with the maximum operating current, the inductor is then physically designed. In addition, it is preferred to construct the load inductor by connecting several small equivalent inductors in series. Each inductor has only one layer of windings to minimize its equivalent parallel capacitance (EPC). WBG devices with fast switching speed are highly sensitive to parasitics, so load inductor EPC minimization is critical.

The dc capacitor is employed to support a stable dc bus voltage across the phase-leg. One bank of bulky capacitors supplies energy to establish the inductive current during the first pulse; while another bank of decoupling capacitors with high current capability and minimal equivalent series inductance is responsible for supplying transient current during the switching transition.

The capacitance of the dc capacitor bank $C_{\rm bulk}$ is selected to limit the bus voltage variation $\Delta V_{\rm DC}$ when the energy transfers from $C_{\text{bulk}}$ to $L$ during the first pulse, that is

$$C_{\text{bulk}} \geq \frac{L{I_L}^2}{(2V_{\text{DC}} - \Delta V_{\text{DC}})\,\Delta V_{\text{DC}}} = \frac{L{I_L}^2}{(2-k_v)\,k_v {V_{\text{DC}}}^2} \approx \frac{L{I_L}^2}{2k_v {V_{\text{DC}}}^2} \quad (2)$$

where $k_v$ is the voltage ripple coefficient, which is practically selected to be 1–5%. Unlike the load inductance, the worst condition for dc capacitance selection is under the minimum operating voltage and maximum current. Generally, electrolytic capacitor and film capacitor are preferred for the dc capacitor bank $C_{\text{bulk}}$.

The capacitance of the decoupling capacitors $C_{\text{dec}}$ is selected to supply transient current and mitigate overvoltage during the switching transient. According to [25], [26], $C_{\text{dec}}$ is practically chosen as

$$C_{\text{dec}} > 250 \times C_{\text{oss}} \quad (3)$$

where $C_{\text{oss}}$ is the output capacitance of the DUT. Film capacitor and ceramic capacitor are preferred for the decoupling capacitor bank $C_{\text{dec}}$.

The gate driver circuit is another crucial component of the DPT circuit and has been extensively investigated [27], [28]. Its detailed design criteria will not be repeated in this paper. It is noteworthy that to fully utilize the high switching-speed capability of WBG devices, special attention should be given to the driving capability of gate driver IC (e.g., rise/fall time, source/sink current, pull-up/pull-down resistance), common mode transient immunity of the gate driver isolation, and cross-talk suppression in a phase-leg [24], [29].

### B. Layout Design Considerations

The purpose of a DPT is to experimentally gather the current and voltage waveforms of the power devices to evaluate their switching behaviors in an actual power converter. Therefore, the primary guideline for the DPT layout is the similarity between the DPT and the intended converter. Second, the parasitics minimization should be considered.

To easily apply the same layout design as the DPT in future converters, it is useful to separate the power stage from the gate driver and modularize the gate driver for each power device. One power stage motherboard can be designed with two gate driver daughter-boards, each integrated with a power device. In this way, the gate driver layout can be standardized for the future converter regardless of the specific topologies.

Parasitic impedance minimization is the other critical consideration for the DPT layout. As shown in Fig. 3, there are mainly six parasitic parameters in the DPT, including $L_{gs}$ and $C_{gs}$ in the gate loop, $L_{ds}$ and $C_{ds}$ in the power loop, and mutual parasitics $L_{cm}$ and $C_{gd}$ shared by the gate and power loops, [5], [30]–[34]. Among them, power loop inductance $L_{ds}$, common source inductance $L_{cm}$, and Miller capacitance $C_{gd}$ significantly impact the switching performance of fast-switching WBG devices.

Power loop inductance $L_{ds}$ is the biggest contributor to the power loop ringing and overshoot voltage, causing electromagnetic interference and reliability issues [5]. The magnetic field

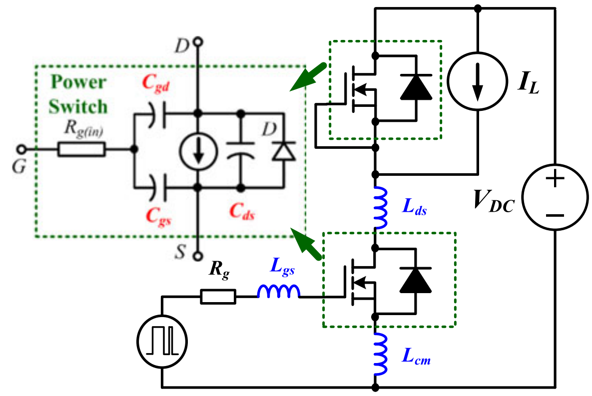


Fig. 3. Six parasitic parameters in a DPT board.

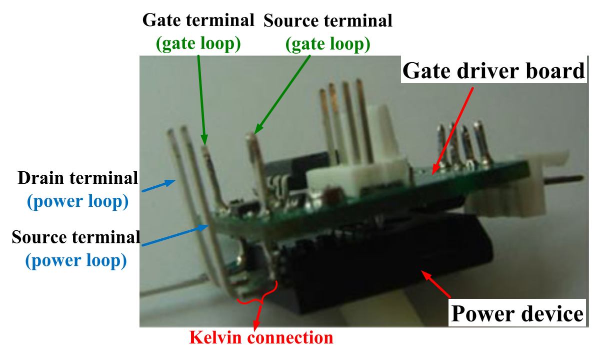


Fig. 4. Modularized design for DPT layout.

cancellation technique and the P-cell and N-cell concept are effective methods to mitigate the power loop parasitic inductance in the PCB layout [10], [12], [35], [36].

In addition, the mutual parasitics $L_{cm}$ and $C_{gd}$ serve as negative feedbacks from the power loop to gate loop during *di/dt* and *dv/dt* transients, leading to slower switching and more switching losses [5]. A modular design of the gate driver with each power device is beneficial for $L_{cm}$ and $C_{gd}$ minimization. Taking one SiC MOSFET in a TO−247 package as an example, a practical example of the layout design is shown in Fig. 4. Separated from the power loop on the motherboard, a standalone gate driver daughterboard is designed for each discrete power device. A Kelvin source connection with separate gate loop and power loop paths minimizes $L_{cm}$. Meanwhile, due to the gate driver daughterboard, the power device has only source and drain terminals connected with the motherboard, such that external parasitic capacitance between drain and gate terminals introduced by print circuit board (PCB) layout can be eliminated to avoid additional impact on internal $C_{gd}$ of the device. Overall, this modular design minimizes the impact of layout parasitics, and is easily applicable to the actual power converter.

## III. Measurement Issues

The high-speed switching transients of the WBG device require the probes to have sufficient bandwidth in order to capture the fast rise and fall edges of the switching waveforms with high fidelity. According to signal theory, the effective bandwidth of a

TABLE I
LATEST VOLTAGE PROBES FOR VDS MEASUREMENT [37], [38]

| Types | Models | Max. voltage | Bandwidth | Rise time* | Pros | Cons |
|---|---|---|---|---|---|---|
| Differential probes | THDP0200 | ± 1600 V | 200 MHz | 8.75 ns | Galvanic isolation | Limited bandwidth, longer connection leads to device |
| | TMDP0200 | ± 750 V | 200 MHz | 8.75 ns | | |
| Passive probes | P5150/P5100A | 2500 V | 500 MHz | 3.50 ns | High bandwidth | Non-galvanic isolation |
| | TPP0850 | 2500 V | 800 MHz | 2.19 ns | | |

* Rise time is calculated based on (4) with 5 times margin (i.e., $5 \times 0.35/\mathrm{Bandwidth}$).

TABLE II
CURRENT MEASUREMENT TECHNIQUES [39]–[42]

| Types/Models | Current type | Bandwidth | Rise time* | Pros | Cons |
|---|---|---|---|---|---|
| Coaxial shunt (0.1 Ω), SSDN-10 | dc/ac | 2000 MHz | 0.88 ns | High bandwidth, high accuracy | Non-galvanic isolation, non-continuous operation |
| Coaxial shunt (0.015 Ω), SSDN-015 | dc/ac | 1200 MHz | 1.47 ns | High bandwidth, high accuracy, >300 A current measurement | Non-galvanic isolation, non-continuous operation |
| Current transformer, Pearson 2877 | ac | 200 MHz | 8.75 ns | High bandwidth, galvanic isolation | Saturation for large current, non-continuous operation |
| Split core current probe, TCP0030A | dc/ac | 120 MHz | 14.58 ns | Galvanic isolation, continuous operation | Low accuracy, limited bandwidth |
| Rogowski coil, CWT Mini | ac | 17 MHz | 102.9 ns | Galvanic isolation | Low bandwidth and accuracy |

* Rise time is calculated based on (4) with 5 times margin (i.e., $5 \times 0.35/\mathrm{Bandwidth}$).

slope signal with a rise time $t_r$ or a fall time $t_f$ can be expressed as [5], [6]

$$f_{\text{eff}} = \frac{0.35}{\min(t_r, t_f)}. \tag{4}$$

Therefore, based on (4), the maximum equivalent frequency of the measured signal $f_{\text{eff}}$ is determined by the fastest switching transient to be measured. To provide an accurate measurement, in practice, system bandwidth should be three to five times higher than the maximum equivalent frequency of the measured signal for the magnitude consideration, and ten times higher for the phase [6]. In the following minimum rise/fall time estimation (see Tables I and II), five times is employed as the margin. Traditionally, the switching characterization of the power devices includes three waveforms: gate-source voltage $V_{gs}$, drain-source voltage $V_{ds}$, and drain current $I_d$, as displayed in Fig. 1 [5], [6]. $V_{ds}$ and $I_d$ are the two waveforms directly used to calculate switching time, $di/dt$, $dv/dt$ and switching losses.

### A. Voltage Measurement

High bandwidth and high accuracy with sufficient dynamic range are the two requirements of the voltage probes for $V_{ds}$ measurement. Voltage probes can be divided into two main categories: differential probes and passive probes. Table I summarizes the latest voltage probes for 600/1200-V WBG devices, including their specifications and overall characteristics. Passive probes are preferred due to their higher bandwidth and dynamic range.

For $V_{gs}$ measurement, dynamic range is not as challenging as compared to the $V_{ds}$ measurement. For the low-side $V_{gs}$ measurement in the phase-leg configuration, probe bandwidth is the main consideration, which practically is easy to satisfy (e.g., 1-GHz TPP1000 passive probe capable of capturing $<1.75$ ns rise/fall edge). For the high-side $V_{gs}$ measurement, in addition to the probe bandwidth, galvanic isolation, common mode (CM) input voltage range, and CM rejection ratio are critical. Practically, it is difficult to get an optimal tradeoff between bandwidth, CM input voltage range, CM rejection ratio, and probe cost for high-side $V_{gs}$ measurement. For the switching characterization, high-side $V_{gs}$ measurement is good to have (e.g., it is useful to identify cross-talk of the upper switch when the lower switch in the phase-leg is operating), but not necessary.

### B. Current Measurement

Several current measurement techniques exist for $I_d$ measurement, including coaxial shunt, current transformer, split core current probe, and Rogowski coil. The state-of-the-art models of each type and their characteristics are summarized in Table II, including the fastest rise time that can be measured accurately based on the specified bandwidth. Due to its limited bandwidth, the Rogowski coil is typically not suitable for switching current measurement.

In addition to the bandwidth, accuracy is another important criterion for transient current measurement. A comparison test was conducted to compare the measurement accuracy of a coaxial shunt, current transformer, and split core current probe, as shown in Fig. 5(a). The devices for this DPT were CMF20120D SiC MOSFETs with C2D10120D SiC Schottky barrier diodes (SBD). During the switching transient of the upper switch, the switching waveforms of the lower side SBD were monitored. The switching current through the SBD was measured by the coaxial shunt, Pearson current transformer, and split core current probe listed in Table II. Due to the excellent reverse recovery characteristics of the SBD, the switching loss should be nearly zero. Fig. 5(b) displays the switching waveforms and losses of the SBD during the switching transient of the upper MOSFET. The switching currents $I_{\text{sw}}$ measured by different current measurement techniques have slight difference, but their corresponding switching losses are quite different (switching losses measured by coaxial shunt, Pearson, and current probe are $0,\ 10$, and 26 $\mu$J, respectively, illustrated in Fig. 5(b)). Clearly, the

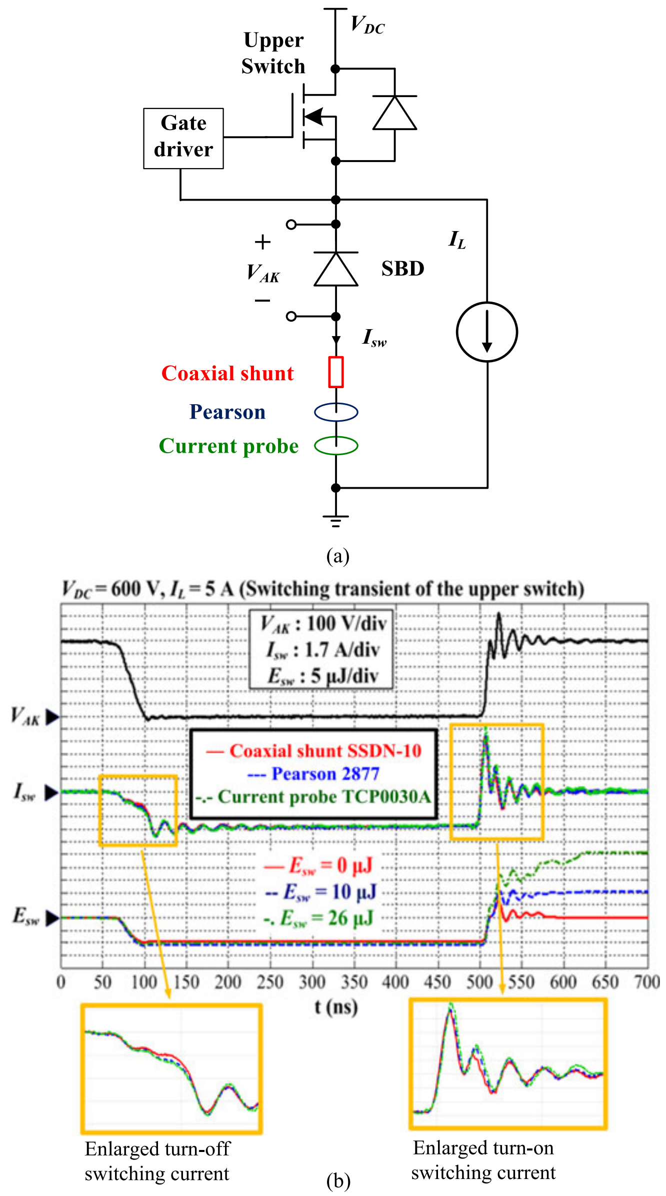


Fig. 5. Accuracy evaluation of different current measurement techniques: (a) test circuit and (b) switching waveforms.

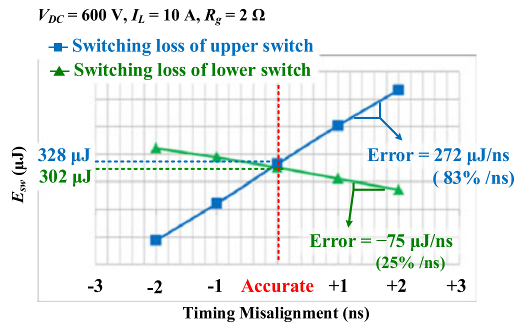


Fig. 6. Sensitivity of V−I alignment.

TABLE III
SENSITIVITY OF V–I ALIGNMENT ON SWITCHING LOSSES

| | | | |
|---|---|---|---|
| $R_g$ ($\Omega$) | 10 | 5 | 2 |
| $E_{\mathrm{sw}\_L}$ ($\mu$J/ ns) | 26 | 132 | 272 |
| $E_{\mathrm{sw}\_H}$ ($\mu$J/ ns) | −16 | −47 | −75 |

accuracy of the switching current measured by the coaxial shunt is the highest, with its measured switching loss close to zero, while the error of the switching loss measured by the current probe is the largest. Thus, the current probe has the lowest accuracy of those tested. This may be explained by the bandwidth of different current measurement techniques. As can be observed in Table II, coaxial shunt has the highest bandwidth, while current probe has the lowest bandwidth. Also, the coaxial shunt $0.015\,\Omega$ SSDN-015 is capable of measuring >300-A current, which can well cover the high current state-of-the-art commercially available WBG power modules [43].

Each current measurement technique will introduce additional parasitic inductance in the power loop, which negatively affects the switching behavior of the DUT. This impact is more serious for the fast WBG power devices with small PCB footprints, such as EPC GaN transistors, since this parasitic inductance may dominate the total power loop inductance. Therefore, the DPT results do not truly represent the switching performance of the DUT in the actual converter. Several methods have been proposed to mitigate this issue, such as a custom current shunt design with tens of $1\,\Omega$ 0201 footprint resistors in parallel to minimize the dimensions of the shunt and its parasitic inductance [1], [44]. However, these small size custom current sensors are not popular concerning their limited bandwidth and measurement accuracy.

### *C. Switching V–I Timing Alignment*

Each of the voltage and current probes has its own characteristic propagation delay. The skew between these two delays will cause inaccurate measurement of switching losses, especially for fast-switching WBG devices.

Following the configuration in Fig. 1, where the lower switch is the operating switch and the upper one in the phase-leg is the synchronous switch, Fig. 6 shows the effect of V−I timing misalignment on the total switching losses of both the lower and upper switches (i.e., $E_{\mathrm{sw}\_L}$ and $E_{\mathrm{sw}\_H}$) in a phase-leg under the operating condition of 600-V/10-A with a gate resistance of $2\,\Omega$. It shows that 1 ns V−I timing misalignment causes 272 $\mu$J error in $E_{\mathrm{sw}\_H}$ and $-75$ $\mu$J error in $E_{\mathrm{sw}\_L}$. Accordingly, 83% error in upper switch switching loss per ns misalignment, 25% error per ns with respect to the lower switching loss, and 31% error of total switching loss per ns will be introduced. More experimental data under different gate resistances are summarized in Table III. Thus, V−I alignment is a critical factor in the accuracy of switching loss evaluation for WBG devices.

There are several recommendations to minimize V-I misalignment:

1) For differential voltage probes and active current probes, the propagation delays are different under different dynamic measurement ranges. Taking the THDP0200 differential voltage probe as an example, the difference in

delay between 150-V dynamic range and 1500-V dynamic range is around 1 ns. Therefore, the active probes should be deskewed under the high dynamic range of the intended experiment.

2) For the coaxial shunt, in addition to the delay induced by the coaxial cable, the shunt itself will incur delay. For instance, with the same coaxial cable, the delay of SSDN-10 shunt is around 1.4 ns shorter than that of SSDN-015.
3) Even for probes of the same model, the propagation delays may still be slightly different. Hence, the oscilloscope channel should be deskewed whenever the probe is changed.

Four methods for V−I alignment are described below, with their pros and cons:

1) Apply a power measurement deskew and calibration fixture [45]. This method is convenient, but less compatible due to the connector constraints. Only current probes from Tektronix can be deskewed with this fixture. Also, its maximum allowable voltage is 8 Vrms, which is not suitable for the calibration of active probes with high dynamic range.
2) Use the probe compensation output of the scope as a standard square waveform signal source for V−I alignment [5]. In addition to the low input voltage (approximately 2.5 V), another issue for this method is that only a coaxial cable rather than the combination of coaxial cable and shunt is deskewed, which will induce timing misalignment in the switching current measurement.
3) Remove the inductor in the DPT and replace one device with a low inductance 100-Ω resistor [8]. In this resistive load DPT, the voltage across a pure resistor is 100× of the current through it, with negligible phase shift induced by the extremely small ratio between series inductance and resistance (approximately 50 ps in practice). The maximum input voltage is determined by the resistor, usually several hundred volts.
4) Use the actual DPT switching waveform based on known V–I relationships [46]. During the *di/dt* of the turn-on transient, a voltage dip can be observed across the drain-source terminals of the operating device due to the parasitic inductance in the power loop. Thus, the current channel deskew can be adjusted on the oscilloscope so that the initial drop in $V_{\mathrm{DS}}$ is graphically aligned to the calculated voltage drop on the power loop inductance (i.e., $L_{ds} \times dI_d/dt$) during the turn-on transient. This method is most effective at high load current due to the longer *di/dt* transient duration. Because this method requires no external dedicated fixture or modification of the test circuit, this method for V−I alignment is preferred. If the magnitude of ringing or noise in the current or voltage channel is comparable to the actual magnitude of either waveform, this method can be difficult to implement. Therefore, it is helpful to adjust the dc offset and scale of both current and voltage channels to fill the entire oscilloscope screen during the *di/dt* transient time interval. This deskew can also be performed during data processing, such as in MATLAB, to verify V–I alignment in previously collected data.

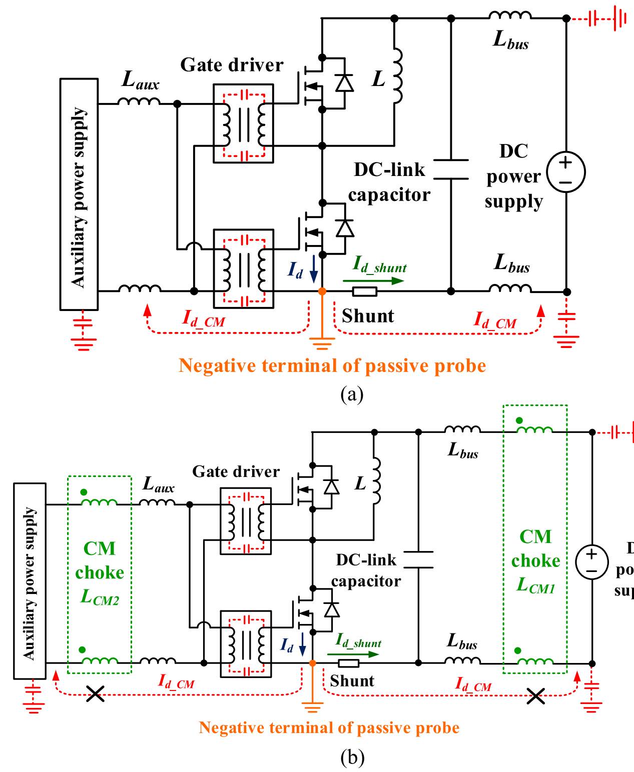


Fig. 7. Capacitively coupled ground loop induced by DPT setup with the consideration of negative terminal of probes (a) CM current due to capacitively coupled ground loop and (b) CM choke for CM current mitigation.

## IV. Probe-Induced Grounding Effects

Considering the bandwidth, dynamic range, and measurement accuracy, passive voltage probes and coaxial shunts have the best performance for switching voltage and current measurements. The negative terminals of both probes' inputs to the oscilloscope are common and tied to the earth ground. The grounding method of these probes will impact the measurement of switching waveforms.

### A. Grounding Loop Effects

Fig. 7 illustrates a DPT setup considering the earth ground. The negative terminal of probes in Fig. 7(a) enables the source terminal of the lower switch to share the same potential with the earth ground. Also, the dc power supplies, including main dc power supply and auxiliary power supply are connected to the DPT board by long wires, contributing to nonnegligible parasitic inductances ($L_{\mathrm{aux}}$ and $L_{\mathrm{bus}}$). Their output terminals create a significant capacitive coupling to the earth ground, due to the copper foil for windings of the isolation transformer used in many power supplies. Therefore, during the switching transient, in addition to the coaxial shunt, the drain current of the lower switch is also bypassed by the ground loop formed by the parasitic inductance and ground coupled capacitance. This extra current induced by the earth ground of the oscilloscope causes ringing in the measured switching current $I_{\mathrm{d}}$, as shown in Fig. 8.

To alleviate this issue, the basic idea is to increase the impedance of the grounding loop, then with the given noise

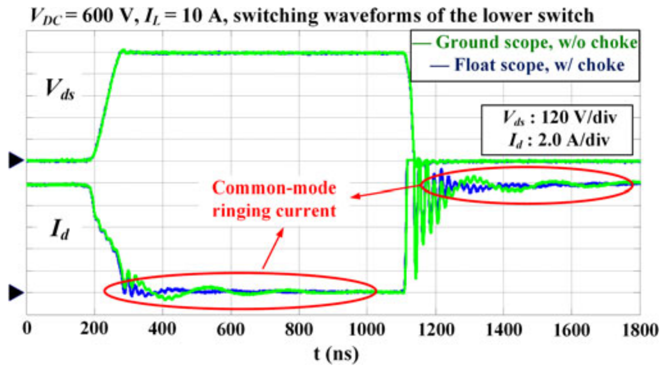


Fig. 8. Impact of capacitive coupled ground loop on drain current measurement.

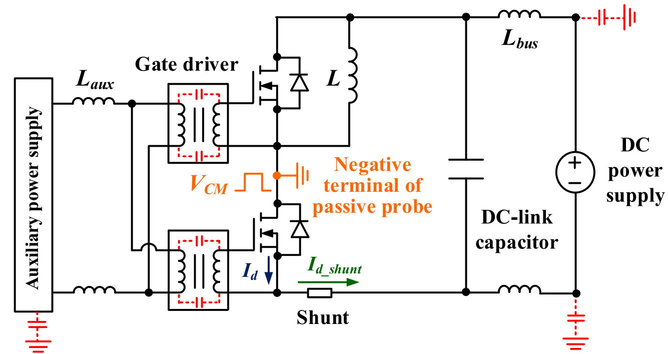


Fig. 9. Capacitive coupled ground loop when probe is attached at the middle point of the phase-leg configuration.

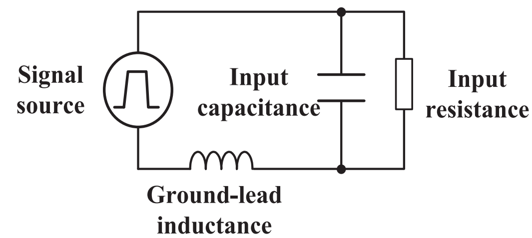


Fig. 10. Equivalent circuit of a typical passive probe considering grounding-lead.

voltage, the noise current will be suppressed. CM chokes with high impedance at the CM ringing frequency can be employed on the main dc power supply and auxiliary power supply, as shown in Fig. 7(b). This ensures very little CM current will flow through the capacitance to the ground as a result of the high impedance of the chokes [6]. These chokes can be added to the DPT probe cables easily with clip-on ferrite cores. The CM choke $L_{\mathrm{CM1}}$ is inserted between the dc power supply and dc-link capacitor in the DPT board, and its introduced leakage inductance will be decoupled by the dc-link capacitor and will not affect the switching performance. Another solution is to isolate the oscilloscope. With a floating ground, the capacitively coupled grounding effect becomes less, and the corresponding CM current will be suppressed. The operator must be cautious about the potential safety hazard that may be introduced by the isolated oscilloscope. Also, accuracy is a concern from the oscilloscope manufacturer when the oscilloscope is floated. Fig. 8 illustrates the comparison of the measured drain current during switching transients with a grounded and isolated oscilloscope. It shows that the CM ringing current of drain current measurement induced by grounding loop effects is suppressed by floating the scope and employing CM chokes. In practice, concerning the safety and accuracy of a floating oscilloscope, employing the CM choke to suppress the grounding loop effect and CM current is preferred.

An isolated dc power supply is assumed for the aforementioned discussion. In this case, the ground terminals of the coaxial shunt and $V_{ds}$ probe are connected with the lower switch source terminal. The measured voltage across the coaxial shunt (i.e., drain current) has a negative polarity. If a ground-referenced dc power supply (i.e., negative terminal of the dc power supply is tied to the ground) is employed, it is suggested to connect the ground terminals of the coaxial shunt and $V_{ds}$ probe with the dc power supply negative terminal to avoid bypassing the coaxial shunt by the grounds of scope and dc power supply. In this case, the voltage across the coaxial shunt will be measured by $V_{ds}$ probe. But it should be relatively low as compared with the drain-source voltage across the device. Thus, the measurement accuracy should be adequate.

The ground point of probes in the DPT circuit cannot be the middle point in a phase leg, since the voltage fluctuation is very large at this point and will cause large grounding current ringing, as shown in Fig. 9. It is recommended to connect the negative terminal of the probes to the dc bus to minimize this ringing.

### *B. Grounding-Lead Inductance Effects*

Another factor impacting the measurement accuracy of switching waveforms is the grounding-lead inductance. A short grounding lead is typically used with passive probes to make a ground connection. However, there is an inductance associated with the grounding lead, as shown in the equivalent circuit of a passive probe in Fig. 10.

The grounding-lead inductance forms a series resonant network with the input capacitance of the probe. Once this series resonant circuit experiences a pulse, ringing is induced. In addition, excessive ground-lead inductance will suppress the charging current to the input capacitance and limit the rise time of the pulse signal. To reduce the ground-lead inductance, a PCB probe-tip adaptor is recommended.

## V. Data Processing

Traditionally, only $V_{ds}$ and $I_d$ of the operating device are measured and analyzed to assess the switching loss [5]. However, considering the impact of cross-talk on the switching behavior, the aforementioned two parameters may be insufficient in a high-speed switching test.

### *A. Impact of Cross-Talk on Switching Losses*

In a phase-leg configuration, the high *dv/dt* during the turn-on transient of one device will generate a displacement current through the Miller capacitance of its complementary device. This displacement current flows into the gate loop, leading to a

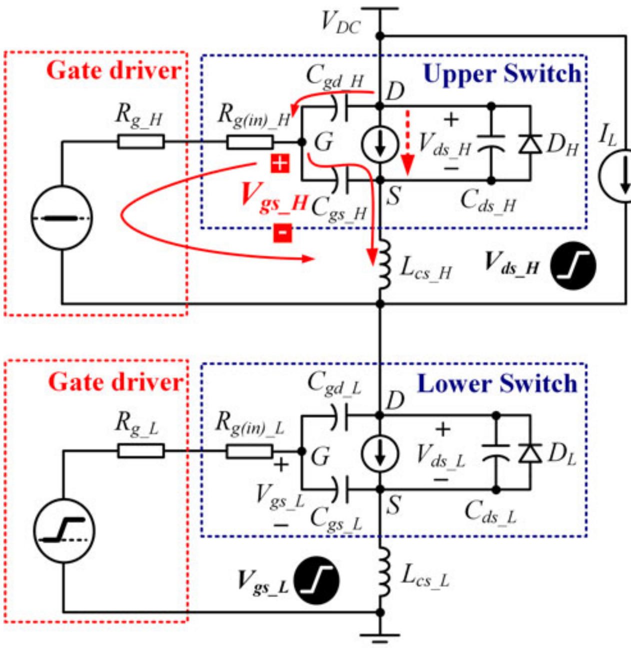


Fig. 11. Mechanism causing cross-talk during the turn-on transient.

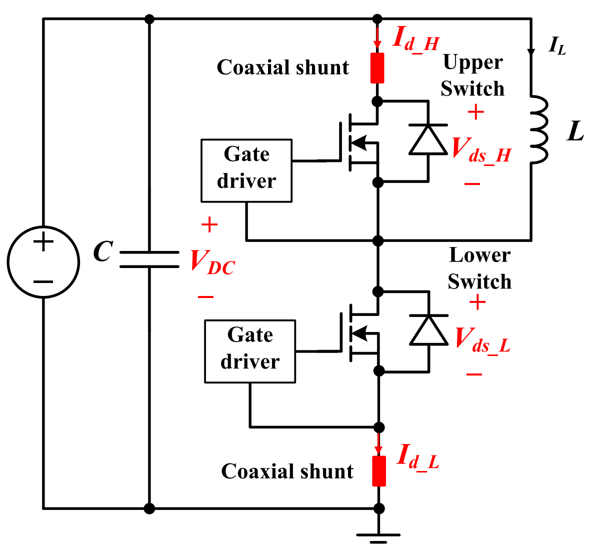


Fig. 12. Measurement parameters in DPT for switching loss evaluation.

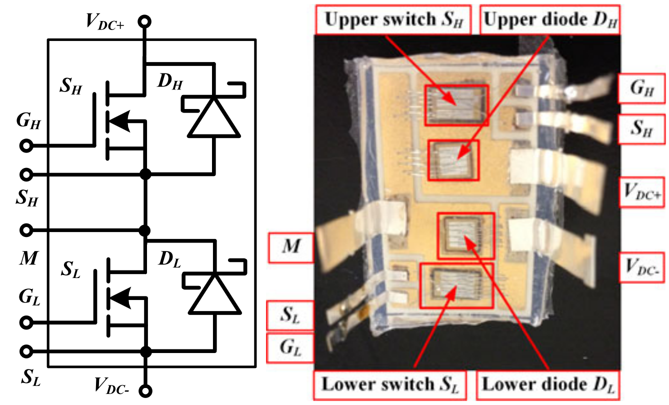


Fig. 13. Diagram of 1200-V SiC-based phase-leg power module.

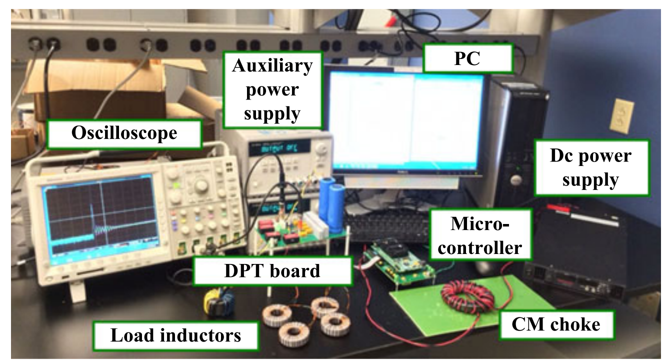


Fig. 14. DPT hardware setup.

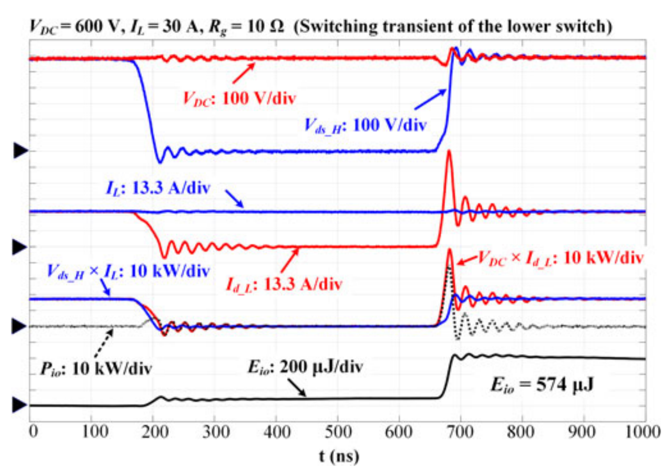


Fig. 15. Switching waveforms for the proposed method.

positive spurious gate voltage, as shown in Fig. 11. This positive spurious gate voltage may begin to turn on the complementary device channel and cause a shoot-through current to flow through both device channels from the dc bus. This shoot-through current can generate significant cross-talk loss in both devices. Therefore, it is possible that during the switching transient of one device, the switching losses of its complementary device become comparable to that of the operating one.

Cross-talk worsens the performance of a power conversion system and should be eliminated in practical applications. But for emerging WBG devices, considering its fast switching and high *dv/dt*, and relatively low threshold voltage, cross-talk is quite easy to occur [47], [48]. Currently, cross-talk suppression techniques for the fast switching WBG devices are developing [47], [49]– [56]. Additionally, although cross-talk exists, it does not necessarily mean a short-circuit failure [24]. Therefore, it is anticipated that with the development of cross-talk suppression techniques for fast switching WBG devices, cross-talk will be eliminated. But at this early stage, it is still good to consider cross-talk and its resultant loss for the switching characterization of emerging WBG devices. It will be helpful to better understand and model the cross-talk so as to suppress the cross-talk in practice.

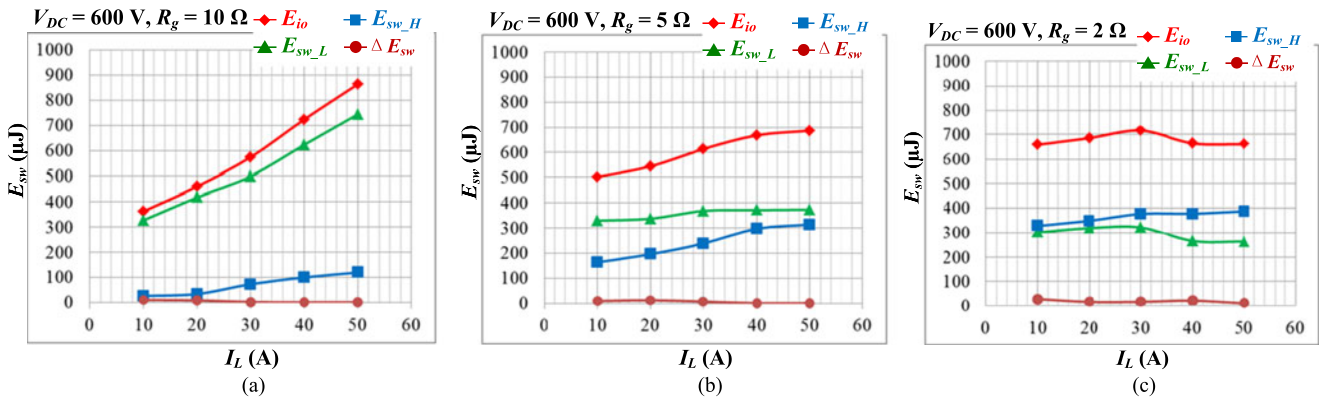


Fig. 16. $E_{io}$, $E_{\mathrm{sw}\_L}$, $E_{\mathrm{sw}\_H}$, and $\Delta E_{\mathrm{sw}}$ dependence on load currents: (a) $R_g = 10\,\Omega$; (b) $R_g = 5\,\Omega$; and (c) $R_g = 2\,\Omega$.

## B. Practical Method for Switching Loss Evaluation

To obtain the total switching losses during the switching transient of the DUT, $V_{ds}$ and $I_d$ of both the upper and lower switch in a phase-leg should be measured. Thus, as shown in Fig. 12, two coaxial shunts should be inserted on the drain terminal of the upper switch and source terminal of the lower switch, which will add more parasitic inductance to the power loop.

A practical method is proposed for comprehensive switching loss evaluation in a phase-leg. According to the conservation of energy, the loss dissipated during the switching transient of the DUT is equal to the difference between the energy supplied by the dc capacitor and the energy stored in the load inductor. Hence, based on measured $I_d$ of the operating device (i.e., current through the dc source), $V_{ds}$ of its complementary device (i.e., voltage across the load inductor), together with the dc bus voltage $V_{\mathrm{DC}}$ and inductive load current $I_L$, the switching losses are given in (5), assuming the lower device to be the device under operation:

$$E_{io} = \int_0^{t_{sw}} \left(V_{\mathrm{DC}} I_{d\_L} - I_L V_{ds\_H}\right) dt \qquad (5)$$

where $E_{io}$ is the total switching loss and $t_{\mathrm{sw}}$ refers to the switching time, including both turn-on and turn-off switching intervals. Energy associated with the inductor including its stored energy and loss are covered in the term $\int (I_L V_{ds\_H}) dt$. Thus, the proposed method only assesses the device related loss excluding the inductor loss.

This method only requires one coaxial shunt, but can still measure the switching losses of both upper and lower devices. Another advantage of this method is its insensitivity to V−I alignment, because each term in (5) includes one transient measurement ($I_{d_L}$ or $V_{ds_H}$) and a relatively constant variable ($V_{\mathrm{DC}}$ or $I_L$).

## C. Experimental Verification

Fig. 13 shows a phase-leg power module built with CPM2-1200-0025B SiC MOSFETs and CPW5-1200-Z050B SiC Schottky diode bare dies from Wolfspeed. A DPT circuit with this phase-leg power module was constructed as shown in Fig. 14. Based on the aforementioned method, Fig. 15 displays

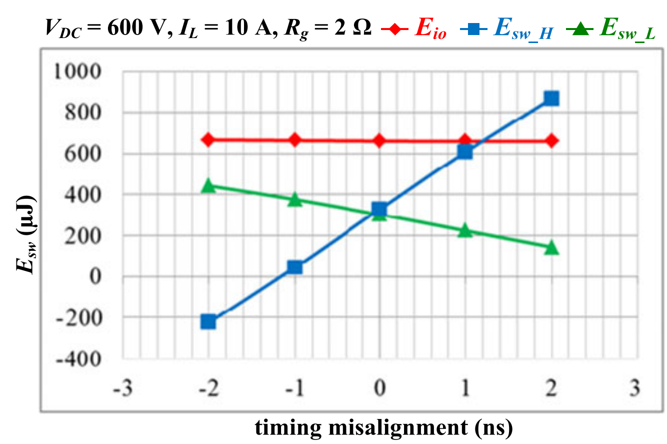


Fig. 17. Comparison of V−I alignment sensitivity among $E_{io}$, $E_{\mathrm{sw}\_L}$, and $E_{\mathrm{sw}\_H}$.

the $V_{\mathrm{DC}}$, $V_{ds\_H}$, $I_L$ and $I_{d\_L}$ during the switching transient of the lower switch under the operating condition of 600-V/30-A with $10 - \Omega$ gate resistance. According to (5), the total switching loss $E_{io}$ is 574 μJ. Using the traditional DPT method, total switching loss of the lower switch $E_{\mathrm{sw}\_L}$ and upper switch $E_{sw\_H}$ under the same operating condition were calculated as 499 and 73 μJ, respectively. Hence, $E_{io}$ is approximately equal to the sum of $E_{\mathrm{sw}\_L}$ and $E_{\mathrm{sw}\_H}$.

More cases with gate resistances of 10, 5, and 2 Ω, operating voltage of 600 V, and operating currents of 10, 20, 30, 40, and 50 A were tested as shown in Fig. 16. $\Delta E_{\mathrm{sw}}$ indicates the difference between $E_{io}$ and the sum of $E_{\mathrm{sw}\_L}$ and $E_{\mathrm{sw}\_H}$. It is shown that $\Delta E_{\mathrm{sw}}$ is nearly zero under any of these operating conditions, which means that $E_{io}$ can accurately evaluate the total switching losses. Due to the cross-talk, $E_{\mathrm{sw}\_H}$ is comparable to $E_{\mathrm{sw}\_L}$ during the switching transient of the lower switch, especially under the operating conditions with small gate resistance. When gate resistance becomes $2\,\Omega$, $E_{\mathrm{sw}\_H}$ is even larger than $E_{\mathrm{sw}\_L}$. $E_{io}$, $E_{\mathrm{sw}\_L}$ and $E_{\mathrm{sw}\_H}$ are the total switching energy loss (turn-on energy loss plus turn-off energy loss).

Fig. 17 shows the sensitivity of $E_{io}$ to V−I timing misalignment. Under the operating condition of 600-V/10-A with $2 - \Omega$ gate resistance, $E_{\mathrm{sw}\_L}$ and $E_{\mathrm{sw}\_H}$ are significantly affected by misalignment, but $E_{io}$ remains constant when the deskew changes. Consequently, the proposed method is a practical and accurate solution for switching loss evaluation of WBG devices in a phase-leg configuration.

## VI. Conclusion

Five factors must be taken into account for the dynamic characterization of WBG devices. First, layout of the DPT board should match that of the actual intended converter and be optimized for the sensitive parasitics (i.e., power loop inductance, common source inductance, and Miller capacitance). Second, probes measuring switching waveforms should satisfy bandwidth, dynamic range, and accuracy requirements. Based on the comparison of the latest probes, high-voltage passive probes and coaxial shunts are preferred. Third, voltage and current channels must be carefully deskewed due to the sensitivity of switching losses to V–I timing misalignment. Four methods of deskew and some general recommendations were summarized. Fourth, probe grounding effects should be suppressed for the accurate measurement of high-speed switching waveforms, using CM chokes and probe-tip adaptors. Fifth, in a phase-leg, the switching losses of the nonoperating devices cannot be ignored due to the shoot-through current from cross-talk, especially for fast-switching WBG devices. To cope with the sensitivity issue of V−I timing alignment and the impact of cross-talk on switching loss evaluation, a practical method for switching loss calculation is proposed, using the difference between the input energy supplied by the dc capacitor and the output energy stored in a load inductor. Based on 1200-V/50-A SiC MOSFET power module, the experimental results show that this method can accurately obtain the switching loss of both devices by detecting only one switching current and voltage, and the results are insensitive to V−I misalignment.

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## References

[1] M. Danilovic *et al.*, “Evaluation of the switching characteristics of a gallium-nitride transistor,” in *Proc. IEEE Energy Convers. Congr. Expo.*, Sep. 17–22, 2011, pp. 2681–2688.

[2] R. Burgos, Z. Chen, D. Boroyevich, and F. Wang, “Design consideration of a fast 0-Ω gate-drive circuit for 1.2kV SiC JFET devices in phase-leg configuration,” in *Proc. IEEE Energy Convers. Congr. Expo.*, Sep. 2009, pp. 2293–2300.

[3] W. Zhang *et al.*, “Evaluation of 600 V cascode GaN HEMT in device characterization and all-GaN-based LLC resonant converter,” in *Proc. IEEE Energy Convers. Congr. Expo.*, Sep. 15–19, 2013, pp. 3571–3578.

[4] I. Josifovic, J. Popovic-Gerber, and J. A. Ferreira, “Improving SiC JFET switching behavior under influence of circuit parasitics,” *IEEE Trans. Power Electron.*, vol. 27, no. 8, pp. 3843–3854, Aug. 2012.

[5] C. Zhen, “Characterization and modeling of high-switching-speed behavior of SiC active devices,” M.S. thesis, Dept. Elect. Eng., Virginia Polytech. Inst. State Univ., Blacksburg, VA, USA, 2009.

[6] J. B. Witcher, “Methodology for switching characterization of power devices and modules,” M.S. thesis, , Dept. Elect. Eng, Virginia Polytech. Inst. State Univ., Blacksburg, VA, USA, 2003.

[7] CREE CPWR AN-08. Application considerations for silicon carbide MOSFETs, 2011. [Online] Available: http://www.cree.com

[8] B. Callanan, “SiC MOSFET double pulse fixture,” 2011. [Online] Available: http://www.cree.com

[9] Semisouth AN-SS1. Silicon carbide enhancement-mode junction field effect transistor and recommendations for use, 2011. [Online]. Available: http://www.semisouth.com

[10] CREE, “Design considerations for designing with CREE SiC Modules: Part 2. techniques for minimizing parasitic inductance,” 2013. [Online]. Available: http://www.cree.com

[11] J. Strydom, “The eGAN FET-silicon power shoot-out: 2: Drivers, layout,” *Power Electron. Technol.*, vol. 37, pp. 14–19, Jan. 2011.

[12] D. Reusch and J. Strydom, “Understanding the effect of PCB layout on circuit performance in a high frequency gallium nitride based point of load converter,” in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, Mar. 2013, pp. 649–655.

[13] Tektronix Inc.: ABCs of probes—primer., Applicat. Note, 2005. [Online] Available: http://www.tektronix.com.

[14] Tektronix Inc.: Power supply measurement and analysis, Applicat. Note, 2011. [Online] Available: http://www.tektronix.com.

[15] Tektronix Inc.: Improve measurement accuracy and reduce cost with passive probes from Tektronix, Appl. 2011. [Online] Available: http://www.tektronix.com

[16] Tektronix Inc.: Improving vertical resolution in Tektronix digital phosphor oscilloscopes, Applicat. Note, 2011. [Online] Available: http://www.tektronix.com

[17] S. Xu, W. Sun, and D. Sun, “Analysis and design optimization of brushless DC motor’s driving circuit considering the Cdv/dt induced effect,” in *Proc. IEEE Energy Convers. Congr. Expos.*, Sep. 12-16 2010, pp. 2091–2095.

[18] Q. Zhao and G. Stojcic, “Characterization of Cdv/dt induced power loss in synchronous buck DC-DC converter,” *IEEE Trans. Power Electron.*, vol. 22, no. 4, pp. 1508–1513, Jul. 2007.

[19] R. K. Burra, S. K. Mazumder, and R. Huang, “dv/dt related spurious gate turn-on of bidirectional switches in a high-frequency cycloconverter,” *IEEE Trans. Power Electron.*, vol. 20, no. 6, pp. 1237–1243, Nov. 2005.

[20] T. Wu, “Cdv/dt induced turn-on in synchronous buck regulations,” tech. rep., International Rectifier, 2007. [Online] Available: http://www.irf.com.

[21] A. Elbanhawy, “MOSFET susceptibility to cross conduction,” in *Proc. Power Electron. Technol.*, Apr. 2005, pp. 26–33.

[22] Fairchild Semiconductor AN-7019, “Limiting cross-conduction current in synchronous buck converter designs,” 2005. [Online] Available: http://www.fairchildsemi.com/an/AN/AN-7019.pdf

[23] P. Vekataraghavan and B. Baliga, “The dv/dt capability of MOS-gated thyristors,” *IEEE Trans. Power Electron.*, vol. 13, no. 4, pp. 660–666, Jul. 1998.

[24] Z. Zhang, W. Zhang, F. Wang, L. M. Tolbert, and B. J. Blalock, “Analysis of the switching speed limitation of wide band-gap devices in a phase-leg configuration,” in *Proc. IEEE Energy Conver. Congr. Expo.*, Sep. 15–20, 2012, pp. 3950–3955.

[25] C. Zheng, D. Boroyevich, P. Mattavelli, and K. Ngo, “A frequency-domain study on the effect of DC-link decoupling capacitors,” in *Proc. IEEE Energy Convers. Congr. Expo.*, 2013, pp. 1886–1893.

[26] L. Qian, W. Shuo, A. C. Baisden, W. Fei, and D. Boroyevich, “EMI suppression in voltage source converters by utilizing dc-link decoupling capacitors,” *IEEE Trans. Power Electron.*, vol. 22, no. 4, pp. 1417–1428, Jul. 2007.

[27] P. J. Grbovic, “Advanced power semiconductors: Art of control, from theory to practice,” in *Proc. IEEE Energy Convers. Congr. Expo.*, Denver, CO, USA, 2013, pp. 1–110.

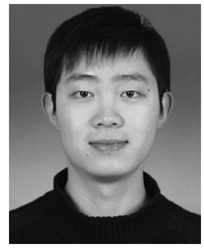
[28] Z. Zhang and F. Wang, “Driving and Characterization of wide bandgap semiconductors for voltage source converter applications,” in *Proc. IEEE WorkshopWide Bandgap Power Devices Appl.*, 2014, pp. 1–84.

[29] Z. Zhang, F. Wang, L. M. Tolbert, B. J. Blalock, and D. J. Costinett, “Understanding the limitations and impact factors of wide bandgap devices’ high switching-speed capability in a voltage source converter,” in *Proc. IEEE Workshop Wide Bandgap Power DevicesAppl.*, Oct. 2014, pp. 7–12.

[30] F. Merienne, J. Roudet, and JL. Schanen, “Switching disturbance due to source inductance for a power MOSFET: Analysis and solutions,” in *Proc. IEEE Power Electron. Spec. Conf.*, 1996, pp. 1743–1747.

[31] Y. Xiao, H. Shah, T. P. Chow, and R. J. Gutmann, “Analytical modeling and experimental evaluation of interconnect parasitic inductance on MOSFET switching characteristics,” in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, 2004, pp. 516–521.

[32] Z. Chen, D. Boroyevich, and R. Burgos, "Experimental parametric study of the parasitic inductance influence on MOSFET switching characteristics," in *Proc. IEEE Int. Power Electron. Conf.*, Jun. 2010, pp. 164–169.
[33] W. Teulings, J. L. Schanen, and J. Roudet, "MOSFET switching behavior under influence of PCB stray inductance," in *Proc. IEEE Ind. Appl. Soc. Annu. Meeting*, 1996, pp. 1449–1453.
[34] H. Zhu, A. R. Hefner, and J. S. Lai, "Characterization of power electronics system interconnect parasitics using time domain reflectometry," *IEEE Trans. Power Electron.*, vol. 14, no. 4, pp. 622–628, Jul. 1999.
[35] S. Li, "Packaging design of IGBT power module using novel switching cells," Ph.D. dissertation, Univ. Tennessee, Knoxville, TN, USA, 2011.
[36] C. Zheng, M. Danilovic, D. Boroyevich, and S. Zhiyu, "Modularized design consideration of a general-purpose, high-speed phase-leg PEBB based on SiC MOSFETs," in *Proc. Eur. Conf. Power Electron. Appl.*, 2011, pp. 1–10.
[37] Tektronix, high-voltage differential probes, Apr. 2013. [Online] Available: http://www.tek.com
[38] Tektronix, passive high voltage probes, Apr. 2013. [Online] Available: http://www.tek.com
[39] PEARSON Current monitor 2877 Datasheet, 2007. [Online] Available: http://www.pearsonelectronics.com/pdf/2877.pdf.
[40] PEM CWT mini specification, Nov. 2010. [Online] Available: http://www.pemuk.com/Userfiles/CWTmini/cwt_mini_1110.pdf.
[41] T & M research products, SSDN series coaxial shunt, 2015. [Online] Available: http://www.tandmresearch.com.
[42] Tektronix, TCP0030A datasheet, Apr. 2013. [Online] Available: http://www.tek.com
[43] Wolfspeed, CAS325M12HM2, 2016. [Online] Available: http://www.wolfspeed.com/cas325m12hm2
[44] J. A. Ferreira, W. A. Cronje, W. A. Relihan, "Integration of high frequency current shunts in power electronic circuits," *IEEE Trans. Power Electron.*, vol. 10, no. 1, pp. 32–37, Jan. 1995.
[45] Tektronix, power measurement deskew & calibration fixture instructions, 2008. [Online] Available: http://www.tek.com
[46] E. Jones *et al.*, "Characterization of an enhancement-mode 650-V GaN HFET," in *Proc. IEEE Energy Convers. Congr. Expo*., Sep. 2015, pp. 400–407.
[47] Z. Zhang, F. Wang, L. M. Tolbert, and B. J. Blalock, "Active gate driver for cross talk suppression of SiC power devices in a phase-leg configuration," *IEEE Trans. Power Electron.*, vol. 29, no. 4, pp. 1986–1997, Apr. 2014.
[48] E. A. Jones, F. Wang, D. Costinett, Z. Zhang, and B. Guo, "Cross conduction analysis for enhancement-mode 650-V GaN HFETs in a phase-leg topology," in *Proc. IEEE Wide Bandgap Power DevicesAppl.*, 2015, pp. 98–103.
[49] J. Wang and H. S. H. Chung, "A novel RCD level shifter for elimination of spurious turn-on in the bridge-leg configuration," *IEEE Trans. Power Electron.*, vol. 30, no. 5, pp. 976–984, Feb. 2015.
[50] Y. Zushi, S. Sato, K. Matsui, Y. Murakami, and S. Tanimoto, "A novel gate assist circuit for quick and stable driving of SiC-JFETs in a 3-phase inverter," in *Proc. IEEE Appl. Power Electron. Conf. Expo*., 2012, pp. 1734–1739.
[51] Q. Zhou, F. Gao, and T. Jiang, "A gate driver of SiC MOSFET with passive triggered auxiliary transistor in a phase-leg configuration," in *Proc. IEEE Energy Convers. Congr. Expo*., 2015, pp. 7023–7030.
[52] E. Velander, A. Löfgren, K. Kretschmar, and H. P. Nee, "Novel solutions for suppressing parasitic turn-on behaviour on lateral vertical JFETs," in *Proc. Eur. Conf. Power Electron. Appl.*, Lappeenranta, Finland, 2014, pp. 1–8.
[53] Q. Zhou and F. Gao, "A gate driver of SiC MOSFET for suppressing the negative voltage spikes in a bridge circuit," in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, 2016, pp. 536–543.
[54] S. Yin *et al.*, "A novel gate assisted circuit to reduce switching loss and eliminate shoot-through in SiC half bridge configuration," in *Proc. IEEE Appl. Power Electron. Conf. Expo*., 2016, pp. 3058–3064.
[55] Rohm Semiconductor, BM60015FV-LB, Jul. 2016. [Online] Available: http://www.rohm.com/web/global/products/-/product/BM60015FV-LB.
[56] STMicroelectronics, "How to fine tune your SiC MOSFET gate driver to minimize losses," Apr. 2015. [Online] Available: www.st.com/resource/en/application_note/dm00170577.pdf.
[57] Z. Zhang *et al.*, "Methodology for switching characterization evaluation of Wide Band-Gap devices in a phase-leg configuration," in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, Mar. 2014, pp. 2534–2541.



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